

On the Goldwasser-Sipser Lower-Bound Protocol

Agni Datta*

University of Edinburgh, UK
agnidatta.org@gmail.com

November 2024

Abstract. We give a self-contained account of the Goldwasser-Sipser lower-bound protocol. The protocol is a public-coin interactive proof for proving that an efficiently recognizable finite set is large. Its mechanism is hashing, in which Arthur chooses a random pairwise-independent hash function and a random target, and Merlin must exhibit a set element that hashes to the target. If the set has size N , then a random target is hit with noticeable probability; if the set has size at most $N/2$, then the hit probability is separated by a constant factor. We place this protocol in the early development of interactive proofs, public coins, approximate counting, universal hashing, and graph isomorphism [WC81, Sto83, Bab85, GMR89, GS86, BL83, Bab16, Bab19]. We prove the hashing lemma carefully, derive the two-round Arthur-Merlin protocol, and explain how graph non-isomorphism is reduced to this set-size lower bound.

1 Introduction

The private-coin proof for graph non-isomorphism is among the cleanest examples of interaction. It sits at the point where the original theory of interactive proofs, developed by Goldwasser, Micali, and Rackoff, meets Babai's public-coin Arthur-Merlin model [Bab85, GMR89]. The verifier hides one random bit by sending a randomly permuted copy of one of two input graphs, and the prover must identify which graph was used. If the graphs are non-isomorphic, an all-powerful prover can answer perfectly. If the graphs are isomorphic, the message distribution is independent of the hidden bit, and the prover can do no better than guess.

This protocol also exposes a limitation. Its soundness depends on the verifier's ability to keep coins private. Arthur-Merlin protocols remove precisely this ability, since Arthur's messages are public random coins. The Goldwasser-Sipser protocol explains why this loss is often not fatal [GS86]. Instead of asking Merlin to recover a hidden random choice, Arthur asks Merlin to prove that a certain set is large. Public randomness selects a random test of largeness, and Merlin can pass the test exactly when the set has enough elements.

The key conceptual move is to replace hidden randomness by a counting gap. In graph non-isomorphism, the private-coin verifier's first-message distribution has twice as many preimages when the input graphs are non-isomorphic as when they are isomorphic. The public-coin protocol does not hide which test was chosen. It publicly chooses a random hash bucket and asks Merlin to find a

*Version 0.2. This manuscript is a working note. Corrections, suggestions, and references to related work are welcome.

witness that the relevant set hits that bucket. Large sets hit many buckets; small sets hit fewer buckets.

There is a useful historical way to read the argument. Stockmeyer had already shown that approximate counting could be controlled by randomized reductions [Sto83], while Wegman and Carter had shown that limited-independence hashing was strong enough for set-equality and authentication tasks [WC81]. Goldwasser and Sipser brought these ideas into interactive proofs by making the approximate-counting question itself interactive. Merlin does not estimate the size of a set numerically. He answers a randomly chosen membership challenge whose success probability is governed by the set size.

Goldwasser and Sipser used this idea to prove the much more general inclusion

$$\text{IP}[k] \subseteq \text{AM}[k + 2],$$

showing that private coins can be simulated by public coins at the cost of two additional rounds [GS86]. This note focuses on the lower-bound protocol that drives the transformation. We isolate the set-size problem, prove the hashing estimate, derive the Arthur-Merlin protocol, and then return to graph non-isomorphism.

The narrative should also be distinguished from the later algebraic revolution in interactive proofs. The sum-check methods of Lund, Fortnow, Karloff, and Nisan and Shamir’s proof that $\text{IP} = \text{PSPACE}$ showed that interaction can certify enormously complex algebraic assertions [LFKN90, Sha90]. The Goldwasser-Sipser protocol is smaller in scope but more surgical. It identifies a single obstruction to public coins, namely the absence of hidden verifier randomness, and resolves it by converting hidden randomness into a set-size lower bound.

2 Surrounding Literature

The Goldwasser-Sipser lower-bound protocol is best read as one part of a short but dense period in which randomness, interaction, and counting became tightly linked. Wegman and Carter gave the hashing language that makes the protocol economical [WC81]. Stockmeyer developed approximate counting as a complexity-theoretic object [Sto83]. Babai introduced public-coin proof systems and used group-theoretic randomness as an algorithmic resource [Bab85]. Goldwasser, Micali, and Rackoff formalized interactive proofs in the language of knowledge and simulation [GMR89]. Goldwasser and Sipser then showed that private verifier coins can be made public by certifying suitable set-size lower bounds [GS86].

Graph non-isomorphism enters this story from two directions. On the proof-system side, it gives a private-coin protocol whose soundness visibly uses hidden randomness, and hence it provides a clean test case for the private-coin versus public-coin question [GS86]. On the algorithmic side, graph isomorphism had already been shaped by canonical-labeling and permutation-group methods [BL83, Luk86, Bab86]. Babai’s later quasipolynomial-time graph-isomorphism algorithm and his subsequent canonical-form report changed the algorithmic landscape but did not remove the pedagogical value of the interactive proof for graph non-isomorphism [Bab16, Bab19]. The protocol remains valuable because it isolates a proof-theoretic phenomenon rather than an algorithmic one.

The neighboring theory of zero knowledge shows that the protocol belongs to a broader method for controlling randomness in proofs. Goldreich, Micali, and Wigderson showed that all languages in NP have zero-knowledge proof systems under standard assumptions [GMW86, GMW91]. Blum,

Feldman, and Micali introduced non-interactive zero knowledge in the common-string model [BFM88], while Ben-Or, Goldwasser, Kilian, and Wigderson developed multi-prover interactive proofs without intractability assumptions [BGKW88]. Damgård and Ostrovsky-Venkatesan-Yung later developed interactive hashing as a tool for zero-knowledge protocol design [Dam94, OVY94]. These works are not needed for the proof below, but they mark the same methodological shift in which randomness is treated as an object to be transformed, simulated, hidden, or certified.

The later complexity consequences put the lower-bound protocol in a larger map. Aiello, Goldwasser, and Håstad studied the power of interaction soon after the Goldwasser-Sipser transformation [AGH86]. The algebraic protocols of Lund, Fortnow, Karloff, and Nisan and Shamir's theorem $IP = PSPACE$ showed that interaction reaches far beyond the original graph examples [LFKN90, LFKN92, Sha90]. Babai, Fortnow, and Lund connected multi-prover interaction to nondeterministic exponential time [BFL90]. Valiant and Vazirani's isolation lemma, although not an interactive-proof result, is another early example in which hashing and uniqueness convert a counting difficulty into a probabilistic decision procedure [VV85]. Later work on statistical zero knowledge, public-coin zero knowledge, and delegation shows how persistent these themes became [GO94, BLP93, Bea91, GV98, BG03, PRT13, Mal15, GKR15].

This note uses that literature only for orientation. The proof we give is elementary once the set-size promise problem has been isolated. The reason to keep the citations visible is that the protocol is conceptually overdetermined, since it belongs simultaneously to hashing, approximate counting, public randomness, graph isomorphism, zero knowledge, and the early structure theory of interactive proofs.

3 Arthur-Merlin Protocols

We use the standard convention that a round is a message sent by one party. In an Arthur-Merlin protocol, Arthur is a probabilistic polynomial-time verifier whose messages are public random strings, and Merlin is computationally unbounded. After seeing Merlin's messages and checking whatever witnesses Merlin provides, the verifier outputs either `accept` or `Reject`. This public-coin restriction is the defining feature of Babai's model [Bab85]; general interactive proofs, introduced in the knowledge-complexity framework of Goldwasser, Micali, and Rackoff, permit the verifier to keep private randomness [GMR89].

Definition 3.1 (Arthur-Merlin protocols). For $k \geq 1$, the class $AM[k]$ consists of languages that have k -round public-coin interactive proofs in which Arthur sends the first message. The class $MA[k]$ is defined similarly, except that Merlin sends the first message. Completeness c and soundness s mean that yes-instances are accepted with probability at least c by some prover, whereas no-instances are accepted with probability at most s against every prover. $\lrcorner$

When $k = 2$, the class $AM[2]$ is denoted by AM . Babai proved that the constant-round public-coin hierarchy collapses in the sense that, for every constant $k \geq 2$, one has $AM[k] = AM[2]$ and $MA[k + 1] = AM[2]$ [Bab85]. We shall not need the collapse theorem, but it explains why two-round public-coin protocols are the canonical form.

The difference between MA and AM can be seen by writing the verifier as a predicate, a viewpoint that is particularly useful in the public-coin hierarchy [Bab85, GS86]. In an MA protocol, Merlin's message is chosen before Arthur's randomness is known; in an AM protocol, Merlin may respond after Arthur's random string has been revealed. Thus, for a verifier predicate $\mathcal{V}(x, \pi, r)$, the yes-condition

for MA has the form

$$\exists \pi \Pr_r [\mathcal{V}(x, \pi, r) = 1] \geq c,$$

whereas the yes-condition for AM has the form

$$\Pr_r [\exists \pi \mathcal{V}(x, \pi, r) = 1] \geq c.$$

The order of the quantifiers is the whole point. The Goldwasser-Sipser protocol exploits the stronger order available in AM, since Arthur first reveals the hash function and target, and Merlin then supplies a preimage.

4 The Set-Size Lower-Bound Problem

We now formulate the promise problem that the protocol solves. The input describes a finite universe $\{0, 1\}^m$, a predicate $\mathcal{R}(x, w)$ decidable in polynomial time, and a threshold N . This formulation is a deliberately interactive analogue of approximate counting, in which one only needs to distinguish two multiplicatively separated cases rather than compute the exact set size [Sto83]. The set of interest is

$$\mathcal{S} = \{x \in \{0, 1\}^m \mid \exists w \mathcal{R}(x, w) = 1\}.$$

Merlin is allowed to certify membership in $\mathcal{S}$ by sending both an element x and a witness w .

Definition 4.1 (Set-size lower bound). The promise problem LB distinguishes instances in which $|\mathcal{S}| \geq N$ from instances in which $|\mathcal{S}| \leq N/2$, under the promise that one of these two cases holds. The verifier is given the predicate $\mathcal{R}$, the ambient length m , and the threshold N . $\lrcorner$

The problem is called a lower-bound problem because Merlin is trying to prove that $\mathcal{S}$ has at least N elements. A direct certificate would list many elements, which is too long. The Goldwasser-Sipser protocol asks for only one element, but chooses the request randomly so that a small set is unlikely to have an answer [GS86]. The central compression of the argument is that many hidden witnesses are replaced by one publicly challenged witness.

To make the random request, Arthur needs a hash family. The family should be samplable using polynomially many random bits, and it should behave like a random function on every pair of distinct inputs. Pairwise independence is the right level of randomness because the analysis below uses only first and second moments; this is the same limited-independence philosophy that appears in universal hashing [WC81].

Definition 4.2 (Pairwise-independent hash family). A family $\mathcal{H}_{m,k}$ of functions from $\{0, 1\}^m$ to $\{0, 1\}^k$ is pairwise independent if, for all distinct $x, x' \in \{0, 1\}^m$ and all $y, y' \in \{0, 1\}^k$,

$$\Pr_h [h(x) = y \wedge h(x') = y'] = 2^{-2k},$$

where h is sampled uniformly from $\mathcal{H}_{m,k}$. It is uniform if, for every $x \in \{0, 1\}^m$ and $y \in \{0, 1\}^k$,

$$\Pr_h [h(x) = y] = 2^{-k},$$

where again h is sampled uniformly from $\mathcal{H}_{m,k}$. $\lrcorner$

Pairwise independence is enough for the protocol because the analysis only needs a first-moment term and a collision correction term. Full independence would be wasteful. This economy is one

reason the protocol is genuinely efficient rather than only information-theoretic.

Proposition 4.3 (A finite-field construction). For every m and $k \leq m$, there is a uniform pairwise-independent family $\mathcal{H}_{m,k}$ that can be sampled with $2m$ random bits. $\square$

Proof. The construction uses affine maps over a field and then projects to k output bits. The affine part gives exact pairwise independence over $\mathbb{F}_{2^m}$, while the projection preserves pairwise independence after summing over the ignored coordinates.

Identify $\{0,1\}^m$ with the field $\mathbb{F}_{2^m}$, fix a surjective linear map proj from $\mathbb{F}_{2^m}$ to $\{0,1\}^k$, and define

$$h_{a,b}(x) = \text{proj}[ax + b],$$

where $a, b \in \mathbb{F}_{2^m}$ are sampled independently and uniformly. The family has 2^{2m} functions, so a uniformly random member can be sampled with $2m$ random bits.

We first prove uniformity. Fix $x \in \mathbb{F}_{2^m}$ and $y \in \{0,1\}^k$. For each choice of a , the value $ax + b$ is uniformly distributed over $\mathbb{F}_{2^m}$ as b ranges uniformly over the field. Since proj is surjective and linear, every $y \in \{0,1\}^k$ has exactly 2^{m-k} preimages under proj . Hence,

$$\Pr_{a,b}[h_{a,b}(x) = y] = \frac{2^{m-k}}{2^m} = 2^{-k}.$$

We next prove pairwise independence. Fix distinct $x, x' \in \mathbb{F}_{2^m}$ and targets $y, y' \in \{0,1\}^k$. For each pair $z, z' \in \mathbb{F}_{2^m}$ with $\text{proj}[z] = y$ and $\text{proj}[z'] = y'$, there is exactly one pair (a, b) satisfying

$$ax + b = z \quad \text{and} \quad ax' + b = z'.$$

Indeed, subtracting the two equations gives

$$a(x - x') = z - z',$$

and $x - x' \neq 0$ in the field, so

$$a = (z - z')(x - x')^{-1}.$$

Then $b = z - ax$ is forced. There are 2^{m-k} choices for z and 2^{m-k} choices for z' , so the number of pairs (a, b) producing the target pair (y, y') is 2^{2m-2k} . Dividing by the total number 2^{2m} of choices for (a, b) gives

$$\Pr_{a,b}[h_{a,b}(x) = y \wedge h_{a,b}(x') = y'] = 2^{-2k}.$$

This is the desired pairwise independence. $\square$

5 The Hashing Lemma

The central estimate concerns the probability that a random hash bucket intersects a fixed set. If $\mathcal{S}$ has size on the order of 2^k , then a random bucket has constant expected intersection with $\mathcal{S}$. The lemma below turns this expectation into a constant-probability event using only pairwise independence, following the hashing estimate at the heart of the Goldwasser-Sipser protocol [WC81, GS86].

Lemma 5.1 (Goldwasser-Sipser hashing estimate). Let $\mathcal{S} \subseteq \{0,1\}^m$ and let $\mathcal{H}_{m,k}$ be uniform and

pairwise independent. If $|\mathcal{S}| \leq 2^{k-1}$, then

$$\frac{3}{4} \cdot \frac{|\mathcal{S}|}{2^k} \leq \Pr_{h,y}[\{x \in \mathcal{S} \mid h(x) = y\} \neq \emptyset] \leq \frac{|\mathcal{S}|}{2^k}.$$

Here h is sampled uniformly from $\mathcal{H}_{m,k}$, and y is sampled uniformly from $\{0,1\}^k$. ▮

Proof. The key observation is that a random bucket is hit with probability close to its expected number of preimages from $\mathcal{S}$, provided the expected number is at most one half. The upper bound is the union bound. The lower bound is the same first-moment calculation corrected by subtracting the probability that two distinct elements collide with the target bucket.

Fix $y \in \{0,1\}^k$ for the moment, and define the event

$$E_x = \{h(x) = y\}$$

for each $x \in \mathcal{S}$. Since $\mathcal{H}_{m,k}$ is uniform, we have

$$\Pr_h[E_x] = 2^{-k}$$

for every $x \in \mathcal{S}$. The union bound gives

$$\Pr_h \left[\bigcup_{x \in \mathcal{S}} E_x \right] \leq \sum_{x \in \mathcal{S}} \Pr_h[E_x] = \frac{|\mathcal{S}|}{2^k}.$$

This is the upper bound for fixed y , and averaging over a uniformly random y preserves it.

For the lower bound, we use the first two terms of inclusion-exclusion. Pairwise independence gives, for every pair of distinct elements $x, x' \in \mathcal{S}$,

$$\Pr_h[E_x \cap E_{x'}] = 2^{-2k}.$$

Therefore,

$$\begin{aligned} \Pr_h \left[\bigcup_{x \in \mathcal{S}} E_x \right] &\geq \sum_{x \in \mathcal{S}} \Pr_h[E_x] - \sum_{\{x,x'\} \subseteq \mathcal{S}} \Pr_h[E_x \cap E_{x'}] \\ &= \frac{|\mathcal{S}|}{2^k} - \binom{|\mathcal{S}|}{2} \cdot 2^{-2k} \\ &\geq \frac{|\mathcal{S}|}{2^k} - \frac{1}{2} \cdot \frac{|\mathcal{S}|^2}{2^{2k}} \\ &= \frac{|\mathcal{S}|}{2^k} \left(1 - \frac{1}{2} \cdot \frac{|\mathcal{S}|}{2^k} \right). \end{aligned} \tag{5.1}$$

The hypothesis $|\mathcal{S}| \leq 2^{k-1}$ implies $|\mathcal{S}|/2^k \leq 1/2$. Substituting this bound into Equation (5.1) gives

$$\Pr_h \left[\bigcup_{x \in \mathcal{S}} E_x \right] \geq \frac{|\mathcal{S}|}{2^k} \left(1 - \frac{1}{4} \right) = \frac{3}{4} \cdot \frac{|\mathcal{S}|}{2^k}.$$

The argument was for an arbitrary fixed target y , so averaging over $y \xleftarrow{s} \{0,1\}^k$ gives the same lower

bound for a uniformly random target. □

6 The Goldwasser-Sipser Protocol

We now turn the hashing estimate into a public-coin protocol for LB. The protocol is meaningful when the threshold N is known, and the verifier can check membership in $\mathcal{S}$ from a short witness. Conceptually, this is approximate counting with a prover, in which Arthur samples a bucket and Merlin supplies the local evidence that the bucket is nonempty [Sto83, GS86].

Theorem 6.1 (Goldwasser-Sipser lower-bound protocol). The promise problem LB has a two-round Arthur-Merlin verifier $\mathcal{V}_{\text{LB}}$. On yes-instances, there exists a prover strategy M for which

$$\Pr_{h,y} [\mathcal{V}_{\text{LB}}^M(\mathcal{R}, N; h, y) = 1] \geq \frac{3}{16}.$$

On no-instances, every prover strategy M^* satisfies

$$\Pr_{h,y} [\mathcal{V}_{\text{LB}}^{M^*}(\mathcal{R}, N; h, y) = 1] \leq \frac{1}{4}.$$

The verifier uses a pairwise-independent hash function and checks a membership witness for $\mathcal{S}$. □

Proof. The protocol asks Merlin to prove that a random hash bucket intersects $\mathcal{S}$. If $\mathcal{S}$ is large, the hashing lemma says that a random bucket is hit with probability bounded below by a constant. If $\mathcal{S}$ is small, even an unbounded Merlin cannot manufacture an element in a bucket that misses the set.

Choose k so that

$$2^{k-2} \leq N \leq 2^{k-1}.$$

Such a k exists for every positive integer N . In particular,

$$\frac{1}{4} \leq \frac{N}{2^k} \leq \frac{1}{2}.$$

Arthur samples $h \xleftarrow{\$} \mathcal{H}_{m,k}$ and $y \xleftarrow{\$} \{0,1\}^k$, and sends (h, y) to Merlin. Merlin responds with a pair (x, w) . The verifier $\mathcal{V}_{\text{LB}}$ outputs one precisely when $h(x) = y$ and $\mathcal{R}(x, w) = 1$.

Suppose first that $|\mathcal{S}| \geq N$. If $|\mathcal{S}|$ is larger than 2^{k-1} , fix an arbitrary subset $\mathcal{S}' \subseteq \mathcal{S}$ of size exactly N . Since $N \leq 2^{k-1}$, Lemma 5.1 applies to $\mathcal{S}'$. Whenever $\{x \in \mathcal{S}' \mid h(x) = y\} \neq \emptyset$, the honest prover strategy M chooses x from this set and sends it together with a witness w satisfying $\mathcal{R}(x, w) = 1$. Therefore,

$$\Pr_{h,y} [\mathcal{V}_{\text{LB}}^M(\mathcal{R}, N; h, y) = 1] \geq \Pr_{h,y} [\{x \in \mathcal{S}' \mid h(x) = y\} \neq \emptyset] \geq \frac{3}{4} \cdot \frac{N}{2^k} \geq \frac{3}{16}.$$

Suppose next that $|\mathcal{S}| \leq N/2$. Fix an arbitrary prover strategy M^* . If $\mathcal{V}_{\text{LB}}^{M^*}(\mathcal{R}, N; h, y) = 1$, then the received pair (x, w) satisfies $\mathcal{R}(x, w) = 1$, so $x \in \mathcal{S}$, and $h(x) = y$. Thus the event in which $\mathcal{V}_{\text{LB}}^{M^*}(\mathcal{R}, N; h, y) = 1$ is contained in the event that the random bucket hits $\mathcal{S}$. By the upper bound in Lemma 5.1,

$$\Pr_{h,y} [\mathcal{V}_{\text{LB}}^{M^*}(\mathcal{R}, N; h, y) = 1] \leq \Pr_{h,y} [\{x \in \mathcal{S} \mid h(x) = y\} \neq \emptyset] \leq \frac{|\mathcal{S}|}{2^k} \leq \frac{N}{2^{k+1}} \leq \frac{1}{4}.$$

This proves the stated completeness and soundness bounds. $\square$

The constants in [Theorem 6.1](#) are deliberately unoptimized. Their role is only to create a positive constant gap. Parallel repetition, or the standard perfect-completeness transformation for Arthur-Merlin protocols, can then produce the usual parameters [[Bab85](#), [GS86](#)]. What matters for the application is that a factor-two gap in set size becomes a constant gap in acceptance probability.

7 Graph Non-Isomorphism

We now recover the public-coin proof for graph non-isomorphism, which was one of the motivating examples for the private-coin versus public-coin question [[GS86](#)]. The problem should be distinguished from the algorithmic graph-isomorphism problem, where Babai and Luks developed canonical-labeling methods and Babai later gave a quasipolynomial-time algorithm and a quasipolynomial-time canonical-form report [[BL83](#), [Luk86](#), [Bab16](#), [Bab19](#)]. Here we ask a proof-system question rather than an algorithm-design question. Let G_0 and G_1 be two graphs on the vertex set $[n]$. We write $G_0 \cong G_1$ if they are isomorphic and $G_0 \not\cong G_1$ otherwise. The language **GNI** consists of pairs (G_0, G_1) with $G_0 \not\cong G_1$.

7.1 The Private-Coin Protocol

The private-coin proof clarifies what the public-coin protocol must replace. It is also a canonical example of how interaction can prove membership in a language whose complement has a direct existential certificate, since graph isomorphism has an NP witness while graph non-isomorphism does not have such an evident one.

Private-coin protocol for GNI

1. The verifier samples $b \xleftarrow{\$} \{0, 1\}$ and $\pi \xleftarrow{\$} \mathfrak{S}_n$, and sends the graph $H = \pi(G_b)$ to the prover.
2. The prover sends a bit $\tilde{b}$.
3. The verifier accepts if $\tilde{b} = b$.

If $G_0 \not\cong G_1$, then the two isomorphism classes are disjoint, so the prover can determine whether H is isomorphic to G_0 or to G_1 . The prover therefore recovers b and is accepted with probability one. If $G_0 \cong G_1$, then the distribution of H is the same for $b = 0$ and $b = 1$. The message reveals no information about b , and every prover is accepted with probability exactly $1/2$.

The hidden bit b is essential in this form of the protocol. Once Arthur reveals all his coins, Merlin sees both the chosen graph and the chosen permutation, so the guessing test no longer has soundness. The set-size reduction below extracts the counting gap that the hidden bit created. The Goldwasser-Sipser point is that the information hidden by private coins can sometimes be replaced by a public proof that a related set has many preimages [[GS86](#)].

7.2 The Set Associated with Two Graphs

Define

$$\mathcal{S}(G_0, G_1) = \{(H, \pi) \mid \pi \in \mathfrak{S}_n \text{ and } H = \pi(G_b) \text{ for some } b \in \{0, 1\}\}.$$

Equivalently, $\mathcal{S}(G_0, G_1)$ contains a graph H together with a permutation π certifying that H is a relabeling of one of the two input graphs. Membership is efficiently checkable, because Merlin can supply the bit b and the verifier can test whether $H = \pi(G_b)$.

Lemma 7.1 (The graph-isomorphism counting gap). If $G_0 \not\cong G_1$, then $|\mathcal{S}(G_0, G_1)| = 2n!$. If $G_0 \cong G_1$, then $|\mathcal{S}(G_0, G_1)| = n!$. $\lrcorner$

Proof. The counting gap is the public-coin shadow of the private hidden-bit test. When the two graphs are non-isomorphic, the bit selecting the graph gives two disjoint ways to generate messages. When the two graphs are isomorphic, the two ways generate the same relabeled graphs, and the apparent factor of two disappears.

For each $b \in \{0, 1\}$ and each permutation $\pi \in \mathfrak{S}_n$, form the pair $(\pi(G_b), \pi)$. If $G_0 \not\cong G_1$, then the pairs arising from $b = 0$ and $b = 1$ are disjoint, because the first coordinate is isomorphic to G_0 in the first case and to G_1 in the second case. For a fixed b , distinct permutations give distinct second coordinates, so they give distinct pairs. Hence, the number of pairs is $2|\mathfrak{S}_n| = 2n!$.

If $G_0 \cong G_1$, then both input graphs have the same isomorphism class. In the set $\mathcal{S}(G_0, G_1)$, the bit b is not recorded. A pair is determined exactly by a permutation $\pi \in \mathfrak{S}_n$ and the graph $H = \pi(G_0)$, because any pair produced from G_1 is also produced from G_0 after fixing an isomorphism between G_0 and G_1 . Thus, there is exactly one pair for each $\pi \in \mathfrak{S}_n$, and the set has size $n!$. $\square$

[Lemma 7.1](#) gives a factor-two set-size gap. Taking $N = 2n!$, the yes-case of GNI has $|\mathcal{S}| = N$, while the no-case has $|\mathcal{S}| = N/2$. Applying [Theorem 6.1](#) yields a public-coin interactive proof, and later work shows that graph non-isomorphism remains a useful test case for understanding proof length, randomness, and collapses of the polynomial-time hierarchy [[KvM99](#)].

Corollary 7.2 (Public-coin proof for graph non-isomorphism). The language GNI has a two-round Arthur-Merlin proof system with a constant completeness-soundness gap. $\lrcorner$

Proof. On input (G_0, G_1) , Arthur and Merlin instantiate the lower-bound protocol with the set $\mathcal{S}(G_0, G_1)$ and threshold $N = 2n!$. In the yes-case $G_0 \not\cong G_1$, [Lemma 7.1](#) gives $|\mathcal{S}| = N$, so the completeness part of [Theorem 6.1](#) applies. In the no-case $G_0 \cong G_1$, the same lemma gives $|\mathcal{S}| = N/2$, so the soundness part applies. Membership in $\mathcal{S}(G_0, G_1)$ is verified by checking a supplied bit b and permutation π against the graph component H , which is polynomial time in the input size. $\square$

8 Private Coins versus Public Coins

The graph non-isomorphism example exhibits the general obstruction in private-coin protocols. The verifier's message is a function of its hidden randomness and the transcript so far. A powerful prover may be able to distinguish large fibers of this message map from small fibers. Goldwasser and Sipser showed that Arthur can make this counting problem public by hashing the relevant set of consistent verifier coins [[GS86](#)]. Thus the lower-bound protocol is not only a protocol for one promise problem; it is a reusable public-coin simulation primitive.

Loosely speaking, the public-coin simulation proceeds as follows. At a point where a private-coin verifier would send a message whose distribution depends on hidden coins, Arthur instead asks Merlin to certify that the set of verifier coins consistent with a favorable continuation is large. The lower-bound protocol supplies exactly this certificate using public randomness. The verifier's private uncertainty has been converted into a public set-size lower bound.

Theorem 8.1 (Goldwasser-Sipser). For every $k \geq 1$, every language with a k -round private-coin interactive proof has a public-coin interactive proof with $k + 2$ rounds. In symbols,

$$\text{IP}[k] \subseteq \text{AM}[k + 2].$$

We do not prove [Theorem 8.1](#) in full here. The proof requires tracking transcripts and conditional acceptance probabilities throughout a general private-coin protocol. Nevertheless, the lower-bound protocol above is the main primitive. It is the mechanism by which Arthur can publicly test that a transcript has many consistent private random choices behind it. In the broader history, this simulation result forms one of the bridges between the early cryptographic view of interaction and the later algebraic characterization of interactive proofs [[GMR89](#), [GS86](#), [LFKN90](#), [Sha90](#)].

9 Concluding Perspective

The Goldwasser-Sipser protocol shows that public randomness can test largeness without enumerating a large set. Merlin does not prove largeness by listing elements; he proves it by responding to a random challenge that a small set is unlikely to answer. The proof combines universal hashing, approximate counting, and interactive verification, three ideas that were developing in the same period [[WC81](#), [Sto83](#), [Bab85](#), [GMR89](#), [GS86](#)].

For graph non-isomorphism, this gives a public-coin replacement for the private hidden-bit protocol. More generally, it explains how private randomness in interactive proofs can be exposed without giving away soundness. The price is a set-size lower-bound question, and pairwise-independent hashing is exactly the tool that makes that question interactive and efficient.

Acknowledgements. I thank Matteo Campanelli for asking me to read about this protocol, and for encouraging me to try to write and think about it carefully.

References

- [AGH86] W. AIELLO, S. GOLDWASSER, and J. HÅSTAD, On the power of interaction, in *27th Annual Symposium on Foundations of Computer Science*, IEEE Computer Society Press, Toronto, Ontario, Canada, October 27–29, 1986, pp. 368–379. <https://doi.org/10.1109/SFCS.1986.36>. (Cited on p. 3).
- [Bab85] L. BABAI, Trading group theory for randomness, in *17th Annual ACM Symposium on Theory of Computing*, ACM Press, Providence, RI, USA, May 6–8, 1985, pp. 421–429. <https://doi.org/10.1145/22145.22192>. (Cited on pp. 1, 2, 3, 8, and 10).
- [Bab86] L. BABAI, A Las Vegas-NC algorithm for isomorphism of graphs with bounded multiplicity of eigenvalues, in *27th Annual Symposium on Foundations of Computer Science*, IEEE Computer Society Press, Toronto, Ontario, Canada, October 27–29, 1986, pp. 303–312. <https://doi.org/10.1109/SFCS.1986.1>. (Cited on p. 2).
- [Bab16] L. BABAI, Graph isomorphism in quasipolynomial time [extended abstract], in *48th Annual ACM Symposium on Theory of Computing* (D. WICKS and Y. MANSOUR, eds.), ACM Press, Cambridge, MA, USA, June 18–21, 2016, pp. 684–697. <https://doi.org/10.1145/2897518.2897542>. (Cited on pp. 1, 2, and 8).

- [Bab19] L. BABAI, Canonical form for graphs in quasipolynomial time: preliminary report, in *51st Annual ACM Symposium on Theory of Computing* (M. CHARIKAR and E. COHEN, eds.), ACM Press, Phoenix, AZ, USA, June 23–26, 2019, pp. 1237–1246. <https://doi.org/10.1145/3313276.3316356>. (Cited on pp. 1, 2, and 8).
- [BFL90] L. BABAI, L. FORTNOW, and C. LUND, Non-deterministic exponential time has two-prover interactive protocols, in *31st Annual Symposium on Foundations of Computer Science*, IEEE Computer Society Press, St. Louis, MO, USA, October 22–24, 1990, pp. 16–25. <https://doi.org/10.1109/FSCS.1990.89520>. (Cited on p. 3).
- [BL83] L. BABAI and E. M. LUKS, Canonical labeling of graphs, in *15th Annual ACM Symposium on Theory of Computing*, ACM Press, Boston, MA, USA, April 25–27, 1983, pp. 171–183. <https://doi.org/10.1145/800061.808746>. (Cited on pp. 1, 2, and 8).
- [Bea91] D. BEAVER, Secure multiparty protocols and zero-knowledge proof systems tolerating a faulty minority, *Journal of Cryptology* **4** no. 2 (1991), 75–122. <https://doi.org/10.1007/BF00196771>. (Cited on p. 3).
- [BGKW88] M. BEN-OR, S. GOLDWASSER, J. KILIAN, and A. WIGDERSON, Multi-prover interactive proofs: How to remove intractability assumptions, in *20th Annual ACM Symposium on Theory of Computing*, ACM Press, Chicago, IL, USA, May 2–4, 1988, pp. 113–131. <https://doi.org/10.1145/62212.62223>. (Cited on p. 3).
- [BG03] M. BEN-OR and D. GUTFREUND, Trading help for interaction in statistical zero-knowledge proofs, *Journal of Cryptology* **16** no. 2 (2003), 95–116. <https://doi.org/10.1007/s00145-002-0113-0>. (Cited on p. 3).
- [BFM88] M. BLUM, P. FELDMAN, and S. MICALI, Non-interactive zero-knowledge and its applications (extended abstract), in *20th Annual ACM Symposium on Theory of Computing*, ACM Press, Chicago, IL, USA, May 2–4, 1988, pp. 103–112. <https://doi.org/10.1145/62212.62222>. (Cited on p. 3).
- [BLP93] J. BOYAR, C. LUND, and R. PERALTA, On the communication complexity of zero-knowledge proofs, *Journal of Cryptology* **6** no. 2 (1993), 65–85. <https://doi.org/10.1007/BF02620135>. (Cited on p. 3).
- [Dam94] I. DAMGÅRD, Interactive hashing can simplify zero-knowledge protocol design without computational assumptions (extended abstract), in *Advances in Cryptology – CRYPTO’93* (D. R. STINSON, ed.), *Lecture Notes in Computer Science* **773**, Springer Berlin Heidelberg, Germany, Santa Barbara, CA, USA, August 22–26, 1994, pp. 100–109. https://doi.org/10.1007/3-540-48329-2_9. (Cited on p. 3).
- [GMW86] O. GOLDBREICH, S. MICALI, and A. WIGDERSON, Proofs that yield nothing but their validity and a methodology of cryptographic protocol design (extended abstract), in *27th Annual Symposium on Foundations of Computer Science*, IEEE Computer Society Press, Toronto, Ontario, Canada, October 27–29, 1986, pp. 174–187. <https://doi.org/10.1109/SFCS.1986.47>. (Cited on p. 2).
- [GMW91] O. GOLDBREICH, S. MICALI, and A. WIGDERSON, Proofs that yield nothing but their validity or all languages in NP have zero-knowledge proof systems, *Journal of the ACM* **38** no. 3 (1991), 691–729. <https://doi.org/10.1145/116825.116852>. (Cited on p. 2).
- [GO94] O. GOLDBREICH and Y. OREN, Definitions and properties of zero-knowledge proof systems, *Journal of Cryptology* **7** no. 1 (1994), 1–32. <https://doi.org/10.1007/BF00195207>. (Cited on p. 3).
- [GV98] O. GOLDBREICH and S. VADHAN, Comparing entropies in statistical zero-knowledge with applications to the structure of SZK, Cryptology ePrint Archive, Report 1998/026, 1998. Available at <https://eprint.iacr.org/1998/026>. (Cited on p. 3).
- [GKR15] S. GOLDWASSER, Y. T. KALAI, and G. N. ROTHBLUM, Delegating computation: Interactive proofs for muggles, *Journal of the ACM* **62** no. 4 (2015), 27:1–27:64. <https://doi.org/10.1145/2699436>. (Cited on p. 3).

- [GMR89] S. GOLDWASSER, S. MICALI, and C. RACKOFF, The knowledge complexity of interactive proof systems, *SIAM Journal on Computing* **18** no. 1 (1989), 186–208. (Cited on pp. 1, 2, 3, and 10).
- [GS86] S. GOLDWASSER and M. SIPSER, Private coins versus public coins in interactive proof systems, in *18th Annual ACM Symposium on Theory of Computing*, ACM Press, Berkeley, CA, USA, May 28–30, 1986, pp. 59–68. <https://doi.org/10.1145/12130.12137>. (Cited on pp. 1, 2, 3, 4, 5, 7, 8, 9, and 10).
- [KvM99] A. KLIVANS and D. VAN MELKEBEEK, Graph nonisomorphism has subexponential size proofs unless the polynomial-time hierarchy collapses, in *31st Annual ACM Symposium on Theory of Computing*, ACM Press, Atlanta, GA, USA, May 1–4, 1999, pp. 659–667. <https://doi.org/10.1145/301250.301428>. (Cited on p. 9).
- [Luk86] E. M. LUKS, Parallel algorithms for permutation groups and graph isomorphism, in *27th Annual Symposium on Foundations of Computer Science*, IEEE Computer Society Press, Toronto, Ontario, Canada, October 27–29, 1986, pp. 292–302. <https://doi.org/10.1109/SFCS.1986.39>. (Cited on pp. 2 and 8).
- [LFKN90] C. LUND, L. FORTNOW, H. J. KARLOFF, and N. NISAN, Algebraic methods for interactive proof systems, in *31st Annual Symposium on Foundations of Computer Science*, IEEE Computer Society Press, St. Louis, MO, USA, October 22–24, 1990, pp. 2–10. <https://doi.org/10.1109/SFCS.1990.89518>. (Cited on pp. 2, 3, and 10).
- [LFKN92] C. LUND, L. FORTNOW, H. J. KARLOFF, and N. NISAN, Algebraic methods for interactive proof systems, *Journal of the ACM* **39** no. 4 (1992), 859–868. (Cited on p. 3).
- [Mal15] L. MALKA, How to achieve perfect simulation and a complete problem for non-interactive perfect zero-knowledge, *Journal of Cryptology* **28** no. 3 (2015), 533–550. <https://doi.org/10.1007/s00145-013-9165-6>. (Cited on p. 3).
- [OVY94] R. OSTROVSKY, R. VENKATESAN, and M. YUNG, Interactive hashing simplifies zero-knowledge protocol design, in *Advances in Cryptology – EUROCRYPT’93* (T. HELLESETH, ed.), *Lecture Notes in Computer Science* **765**, Springer Berlin Heidelberg, Germany, Lofthus, Norway, May 23–27, 1994, pp. 267–273. https://doi.org/10.1007/3-540-48285-7_23. (Cited on p. 3).
- [PRT13] R. PASS, A. ROSEN, and W.-L. D. TSENG, Public-coin parallel zero-knowledge for NP, *Journal of Cryptology* **26** no. 1 (2013), 1–10. <https://doi.org/10.1007/s00145-011-9110-5>. (Cited on p. 3).
- [Sha90] A. SHAMIR, $IP=PSPACE$, in *31st Annual Symposium on Foundations of Computer Science*, IEEE Computer Society Press, St. Louis, MO, USA, October 22–24, 1990, pp. 11–15. <https://doi.org/10.1109/SFCS.1990.89519>. (Cited on pp. 2, 3, and 10).
- [Sto83] L. J. STOCKMEYER, The complexity of approximate counting (preliminary version), in *15th Annual ACM Symposium on Theory of Computing*, ACM Press, Boston, MA, USA, April 25–27, 1983, pp. 118–126. <https://doi.org/10.1145/800061.808740>. (Cited on pp. 1, 2, 4, 7, and 10).
- [VV85] L. G. VALIANT and V. V. VAZIRANI, NP is as easy as detecting unique solutions, in *17th Annual ACM Symposium on Theory of Computing*, ACM Press, Providence, RI, USA, May 6–8, 1985, pp. 458–463. <https://doi.org/10.1145/22145.22196>. (Cited on p. 3).
- [WC81] M. N. WEGMAN and L. CARTER, New hash functions and their use in authentication and set equality, *Journal of Computer and System Sciences* **22** (1981), 265–279. (Cited on pp. 1, 2, 4, 5, and 10).